

CSE 471: MACHINE LEARNING

Bayesian Learning

h_1 : 100% cherry,

h_2 : 75% cherry + 25% lime,

h_3 : 50% cherry + 50% lime,

h_4 : 25% cherry + 75% lime,

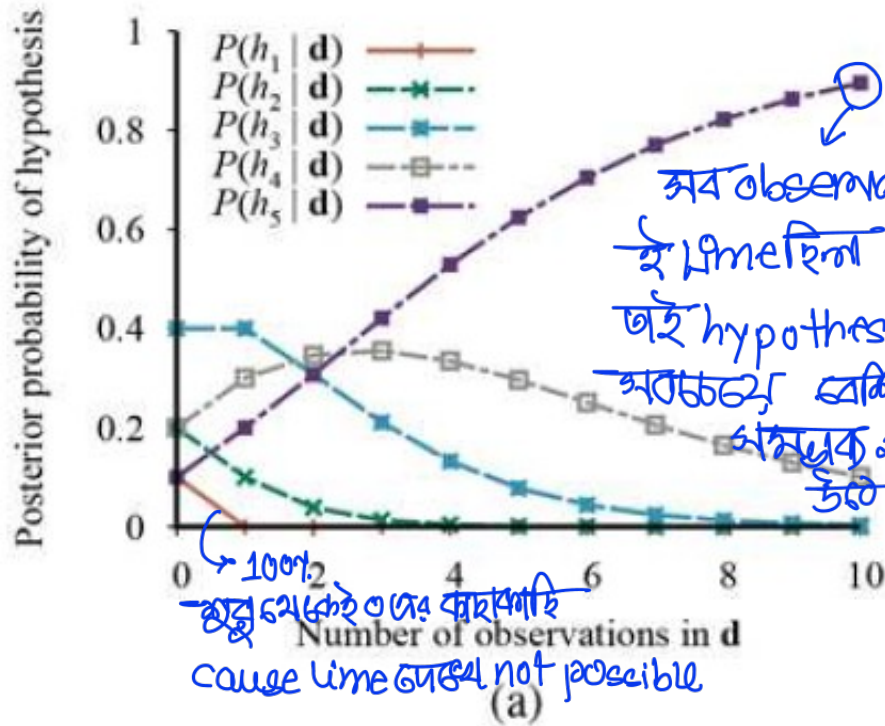
h_5 : 100% lime.

$$P(h_i|\mathbf{d}) = \alpha P(\mathbf{d}|h_i)P(h_i) \leftarrow$$

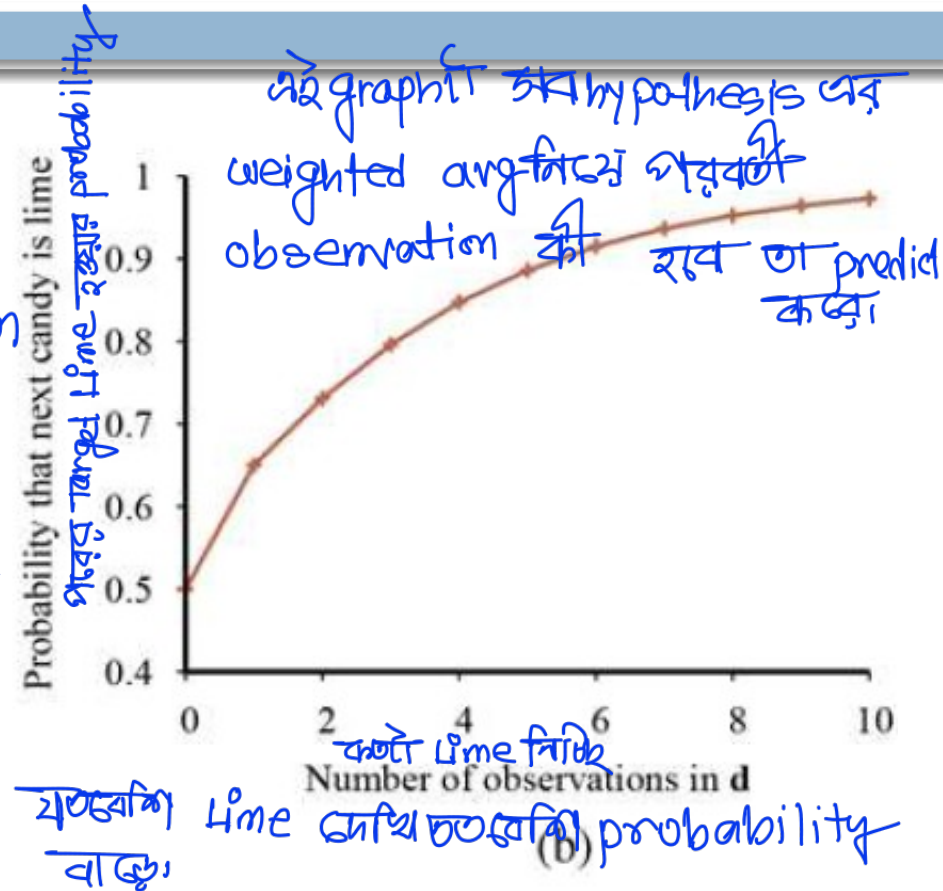
$$\mathbf{P}\left(X|\mathbf{d}\right) = \sum_i \mathbf{P}\left(X|h_i\right)P\left(h_i|\mathbf{d}\right)$$

$$P\left(\mathbf{d}|h_i\right) = \prod_j P\left(d_j|h_i\right)$$

Bayesian Learning



(a) Posterior probabilities $P(h_i|d_1, \dots, d_N)$ from Equation (20.1). The number of observations N ranges from 1 to 10, and each observation is of a lime candy. (b) Bayesian prediction $P(D_{N+1} = \text{lime}|d_1, \dots, d_N)$ from Equation (20.2).



Maximum a posteriori hypothesis

→ *মোট Bayesian learning এ আমরা সব hypo এর weighted sum নিই, কিন্তু complex হিসাব এখন সবচেয়ে সম্ভাব্য একটি hypothesis বের করা*

Single most probable hypothesis

An h_i that maximizes ()

Optimization problem, rather than summation

Often times, it is much easier to solve

MAP → *সবচেয়ে ভালো অনুমান বোঝে নেই, summation না করে ব্লক maximize করে।*

Bayesian	MAP
সব hypo এর sum	ব্লক বো best hypo
কি accurate	কি accurate কিন্তু সহজ
Σ (summation)	optimization (maximization)

$$h_{\text{map}} = \arg \max_{h_i} P(h_i | D)$$

সবচেয়ে বড়

Maximum-Likelihood hypothesis

Uniform prior assumed over the hypothesis space

Find the h that maximizes $(\mid)$

MAPএ prior $p(h_i)$ important ছিল। But MLএ বলা হয় সব hypothesis সমান সম্ভাব্য, তাহলে $p(h_i)$ অবজারগাম্য সমান, তাই বাদ দিতে পারি।

$h_{ML} =$ যে h_i তে $p(d|h_i)$ সবচেয়ে বেশি। (যে $p(d|h_i)$ Likelihood maximize করে)

Bayesian $\rightarrow$ সব hypothesis এর weighted sum maximize করে।
MAP $\Rightarrow p(h_i|d) = \text{prior} \times \text{likelihood}$ কে maximize করে।
ML $\Rightarrow p(d|h_i) = \text{Likelihood}$

ML parameter learning

$$P(\mathbf{d} | h_\theta) = \prod_{j=1}^N P(d_j | h_\theta) = \theta^c \cdot (1 - \theta)^\ell$$

N বার candy উল্লিখ

① c বার cherry
② l বার lime

c চৌ cherry আসে, l টি lime আসার probability

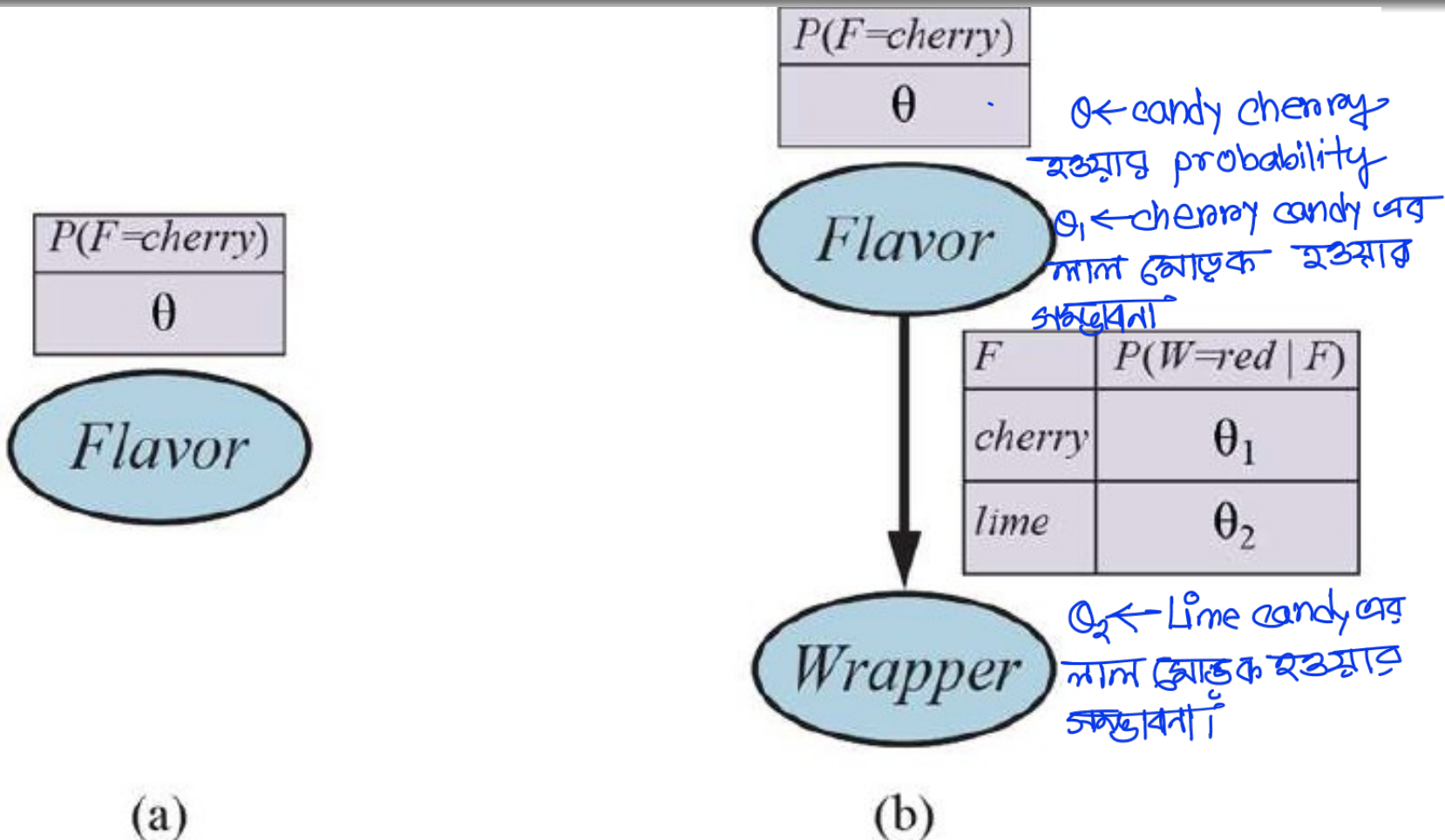
$$L(\mathbf{d} | h_\theta) = \log P(\mathbf{d} | h_\theta) = \sum_{j=1}^N \log P(d_j | h_\theta) = c \log \theta + \ell \log (1 - \theta)$$

Maximize করার জন্য derivative=0 করব

$$\frac{dL(\mathbf{d} | h_\theta)}{d\theta} = \frac{c}{\theta} - \frac{\ell}{1 - \theta} = 0 \quad \Rightarrow \quad \theta = \frac{c}{c + \ell} = \frac{c}{N}$$

→ cherry আসার probability

ML parameter learning



(a) Bayesian network model for the case of candies with an unknown proportion of cherry and lime. (b) Model for the case where the wrapper color depends (probabilistically) on the candy flavor.

ML parameter learning

$$P(\mathbf{d} | h_{\theta, \theta_1, \theta_2}) = \underbrace{\theta^c (1 - \theta)^\ell}_{\substack{\text{c is cherry \& \\ lime being} \\ \text{probability}}} \cdot \underbrace{\theta_1^{r_c} (1 - \theta_1)^{g_c}}_{\substack{\text{cherry candy or not cherry} \\ \text{get green wrapper or not}}} \cdot \theta_2^{r_\ell} (1 - \theta_2)^{g_\ell}$$

$$L = [c \log \theta + \ell \log(1 - \theta)] + [r_c \log \theta_1 + g_c \log(1 - \theta_1)] + [r_\ell \log \theta_2 + g_\ell \log(1 - \theta_2)]$$

$$\frac{\partial L}{\partial \theta} = \frac{c}{\theta} - \frac{\ell}{1 - \theta} = 0 \quad \Rightarrow \quad \theta = \frac{c}{c + \ell}$$

$$\frac{\partial L}{\partial \theta_1} = \frac{r_c}{\theta_1} - \frac{g_c}{1 - \theta_1} = 0 \quad \Rightarrow \quad \theta_1 = \frac{r_c}{r_c + g_c}$$

$$\frac{\partial L}{\partial \theta_2} = \frac{r_\ell}{\theta_2} - \frac{g_\ell}{1 - \theta_2} = 0 \quad \Rightarrow \quad \theta_2 = \frac{r_\ell}{r_\ell + g_\ell}$$

ML parameter learning (Continuous model)

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

একমাত্রিক normal distribution probability density function

$$L = \sum_{j=1}^N \log \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_j-\mu)^2}{2\sigma^2}} = N \left(-\log \sqrt{2\pi} - \log \sigma \right) - \sum_{j=1}^N \frac{(x_j - \mu)^2}{2\sigma^2}$$

$$\frac{\partial L}{\partial \mu} = -\frac{1}{\sigma^2} \sum_{j=1}^N (x_j - \mu) = 0 \quad \Rightarrow \quad \mu = \frac{\sum_j x_j}{N}$$

$$\frac{\partial L}{\partial \sigma} = -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{j=1}^N (x_j - \mu)^2 = 0 \quad \Rightarrow \quad \sigma = \sqrt{\frac{\sum_j (x_j - \mu)^2}{N}}$$

Conditional distribution $P(Y |$

$X)$ → $\rightarrow$ Linear regression ଏବଂ probabilistic distribution

Y has a Gaussian distribution

Fixed standard deviation

Mean is linearly dependent on X

$$P\left(y \mid x\right) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{\left(y - \left(\theta_1 x + \theta_2\right)\right)^2}{2\sigma^2}}$$

→ error ଥିବାର probability ଥିବାର

Linear regression → MSE

$$\left(y - \left(\theta_1 x + \theta_2\right)\right)^2$$

→ actual & predicted ଥିବାର ସମସ୍ତ
→ predictive value

ML parameter Learning



ଅନୁଦାନିନ model ଶିକ୍ଷା



Log Likelihood maximize କରା



Derivative = 0 କରା



ଢେଉ → ଅନୁଗଣନା counting / familiar formula

Linear regression = Gaussian ML

Logistic regression = Bernoulli ML